

TOWARDS A GENERAL HUMANOID LOCO-MANIPULATION MODEL VIA EGOCENTRIC WHOLE-BODY HUMAN DATA PRETRAINING

Anonymous authors

Paper under double-blind review



Figure 1: Introducing λ_0 with **HUMANVERSE-500**. (A) We introduce a robot-free egocentric human-data collection paradigm and use it to collect **HUMANVERSE-500**, a 500-hour dataset of whole-body human loco-manipulation. (B) We develop λ_0 , a unified whole-body humanoid vision-language-action policy, through a three-stage recipe combining large-scale Internet manipulation data, **HUMANVERSE-500**, and embodiment-specific data. (C) Across simulated and real-world loco-manipulation tasks, λ_0 achieves state-of-the-art performance and strong generalization.

ABSTRACT

Humanoid whole-body manipulation has advanced rapidly, enabling policies to coordinate locomotion, posture, bimanual interaction, and dexterous hands over increasingly long horizons. In parallel, egocentric human data have emerged as a scalable source of supervision for manipulation, providing broad coverage of objects, scenes, and everyday activities without requiring robot operation. These two developments do not yet connect cleanly. Existing egocentric action supervision is concentrated on bilateral end-effectors or hand pose, whereas whole-body loco-manipulation additionally depends on global displacement and the temporal coupling between body motion and interaction. Collecting the missing supervision directly through humanoid teleoperation is costly and difficult to scale. We therefore ask how rich human activity can be transferred into a whole-body humanoid foundation policy. We introduce **HUMANVERSE-500**, a 500-hour corpus of target-robot-free first-person video synchronized with wearable-tracked body kinematics and reconstructed bilateral hand motion. A three-stage training recipe produces λ_0 : it first learns broad interaction structure from heterogeneous egocentric datasets, then learns coupled whole-body motion from **HUMANVERSE-500**, and finally grounds the representation with task-aligned human and robot trajectories. Shared vision-language-action representations enable transfer across stages, while domain-specific interfaces preserve the semantics of each stage-action space. We evaluate this transfer through controlled human-data scaling, **SIMPLE**, four real-world loco-manipulation tasks, object-exposure generalization, robot-data-efficiency controls, and stage-wise ablations, measuring both terminal success and ordered task progress.

1 INTRODUCTION

Humanoid loco-manipulation has advanced rapidly as recent vision-language-action (VLA) systems bring lower-body control and bimanual manipulation into a unified language-conditioned policy, despite adopting different action parameterizations, and thereby enable longer-horizon closed-loop behavior (Jiang et al., 2026; Shi et al., 2026; Wei et al., 2026b; Hu et al., 2026). This policy-level unification requires a corresponding whole-body control interface that coordinates locomotion and manipulation during both data collection and execution. SONIC provides such an interface by mapping human motion represented in SMPL to executable humanoid behavior, making unified whole-body teleoperation practical (Luo et al., 2025). Building on SONIC, GR00T N1.6 and ω -0 collect real-robot whole-body demonstrations through the same teleoperation paradigm, with GR00T N1.6 training a VLA policy and ω -0 training a world-action model (WAM) to predict controller-compatible whole-body action representations (GEAR Team et al., 2025; Li et al., 2026b). Together, these developments establish a coherent stack from human whole-body demonstration, through high-level action learning, to coordinated robot execution.

Beyond architectures and control systems, paired action data remains a central bottleneck for VLAs. Unlike language or vision pretraining, action learning requires temporally aligned observations and control targets in a consistent action space. Generic Internet video rarely provides this supervision, historically tying bimanual datasets to robot-in-the-loop teleoperation (Zhao et al., 2023; Fu et al., 2025b; Khazatsky et al., 2024). Portable interfaces decouple collection from target robots while preserving robot-compatible actions (Chi et al., 2024; Xiaomi Robotics Team et al., 2026). Egocentric human video offers a further shift toward natural data collection, provided that action supervision can be measured during capture or reconstructed offline (Kareer et al., 2024; Hoque et al., 2025; Punamiya et al., 2026). Recent pipelines segment long videos into atomic actions, attach language descriptions, recover framewise wrist, hand, and camera motion, and align the resulting human trajectories with robot learning through co-training, retargeting, or visual synthesis (Li et al., 2026a; Yang et al., 2025; Wang et al., 2026). Scaling studies further show that action-labeled egocentric pretraining can improve downstream manipulation and transfer through aligned human-robot training (Zheng et al., 2026b). This progression establishes egocentric video as a practical source of action supervision for robot manipulation.

The supervision that makes ego data effective for manipulation, however, is not yet sufficient for whole-body operation. Most existing action-aligned ego pipelines focus on bilateral wrist trajectories, end-effector motion, or hand pose. These signals describe how a person interacts once an object is within reach, but they omit or weakly constrain the body motion that makes the interaction possible: approaching a workspace, changing root and camera pose, coordinating the torso with a reach, maintaining a grasp during transport, and repositioning for placement. Consequently, simply extending a hand-centric action target does not provide the synchronized whole-body structure required by humanoid loco-manipulation.

Collecting this structure entirely with humanoid teleoperation is an expensive alternative. Whole-body demonstrations require stable hardware, coordinated control of the base, torso, arms, and hands, and repeated recovery from physical failures. Their acquisition rate is therefore tied to robot availability. The scalability of human activity and the specificity of robot control expose the central question of this work: *how can rich egocentric human activity be transferred into a whole-body humanoid foundation policy, while reserving scarce robot data for executable embodiment grounding?*

We study this question with **HUMANVERSE-500**, a 500-hour corpus of target-robot-free first-person activity synchronized with device-tracked whole-body kinematics. Unlike hand-only supervision, each episode preserves global body displacement, bilateral hand geometry, camera transformations, and their native temporal coupling. Collection is distributed across activities rather than concentrated on the evaluation tasks. We separately acquire 600 task-aligned human demonstrations for the four real-world tasks; these are used only during post-training and are excluded from both HUMANVERSE-500 and its scaling axis.

We instantiate this transfer in λ_0 , a whole-body humanoid policy trained in three stages. Stage I uses heterogeneous egocentric datasets to learn broad object and interaction structure. Stage II introduces HUMANVERSE-500 to learn the coupled evolution of approach, reach, manipulation, and transport. Stage III combines the separate task-aligned human set with embodiment-matched robot trajectories to learn executable control. A shared vision-language backbone and generative

action expert carry reusable structure across stages, while human motion and each robot embodiment retain semantically distinct state and action projections. λ_0 therefore shares representations without treating human kinematics as robot actuator commands.

We evaluate whether the resulting human prior improves downstream success, generalization, and robot-data efficiency. Nested 0%, 5%, 10%, 25%, 50%, and 100% subsets of HUMANVERSE-500 isolate human-data scale while holding the model, task-aligned human set, robot trajectories, and post-training budget fixed. Stage-II validation dynamics and the loss-data relationship test whether the learned motion prior scales predictably; SIMPLE and four real-world tasks measure strict terminal success and weighted task progress. Object-exposure splits then distinguish robot-seen, human-only, and fully unseen instances. Together, these experiments test whether scalable whole-body human experience can augment scarce robot supervision and transfer human activity coverage into more capable humanoid loco-manipulation.

2 RELATED WORK

2.1 SCALING ROBOT AND HUMAN EXPERIENCE

Generalist control relies on demonstration diversity across tasks and embodiments (Dasari et al., 2020; Jang et al., 2022; Walke et al., 2023; Open X-Embodiment Collaboration, 2024; Brohan et al., 2023; Zitkovich et al., 2023; Khazatsky et al., 2024; Bu et al., 2025). Shared policies make this diversity reusable (Ghosh et al., 2024; Kim et al., 2025; Chi et al., 2025), while improved teleoperation and portable interfaces reduce acquisition overhead (Dass et al., 2023; Qin et al., 2023; Zhao et al., 2023; Fu et al., 2025b; Chi et al., 2024). These approaches broaden coverage, but robot collection still requires hardware time and physical resets; portable interfaces instead bound the recorded action space. We decouple broad whole-body experience from target-robot availability, using human activity for motion priors and robot data for executable grounding.

2.2 EGOCENTRIC HUMAN DATA AND GEOMETRIC SUPERVISION

Activity-rich video does not automatically supply action-aligned supervision (Grauman et al., 2022; Damen et al., 2022; Goyal et al., 2017; Grauman et al., 2024; Wang et al., 2023b). Temporal and geometric annotations narrow this gap (Liu et al., 2022; Wang et al., 2025; Sener et al., 2022; Ragusa et al., 2021; Huang et al., 2024; Banerjee et al., 2025); tracked or reconstructed demonstrations make human motion directly usable for action learning (Hoque et al., 2025; Punamiya et al., 2026; Li et al., 2026a; Zheng et al., 2026b). Representation learning transfers visual structure (Nair et al., 2023; Sontakke et al., 2023; Kumar et al., 2020), whereas imitation and alignment connect observed behavior to robot control (Shao et al., 2020; Bahl et al., 2022; Wang et al., 2023a; Bharadhwaj et al., 2023; Kareer et al., 2024; Lepert et al., 2025). Hand-object geometry describes local interaction but underdetermines how the body approaches, supports a reach, or transports a grasped object. HUMANVERSE-500 preserves body-hand timing and displacement, supervising their coordination rather than merely adding hand examples. Quality and optimization also matter (Mandlekar et al., 2022); more footage alone need not yield better control.

2.3 HUMAN-TO-HUMANOID TRANSFER AND WHOLE-BODY VLAS

Tracking, retargeting, and teleoperation bridge human motion and robot execution (Luo et al., 2023; Fu et al., 2025a; He et al., 2025; Araujo et al., 2025). Recent humanoid policies exploit human experience through latent video actions, geometric pretraining, or aligned human-robot co-training (Jiang et al., 2026; Wei et al., 2026b; Shi et al., 2026; Hu et al., 2026); predictive and integrated whole-body models further expand these interfaces (Li et al., 2026b; Symbiosis Robotics, 2026; Figure AI, 2026; Parada, 2026; Zheng et al., 2026a). Their common aim is transfer, but broad pretraining and task-matched human demonstrations play different roles: reusable priors versus downstream coverage. Our three stages separate interaction learning, whole-body coordination, and robot grounding. Fixing the model, task-aligned human set, robot data, and adaptation budget tests broad human-data scale without conflating it with extra downstream supervision (Appendix Table 5).

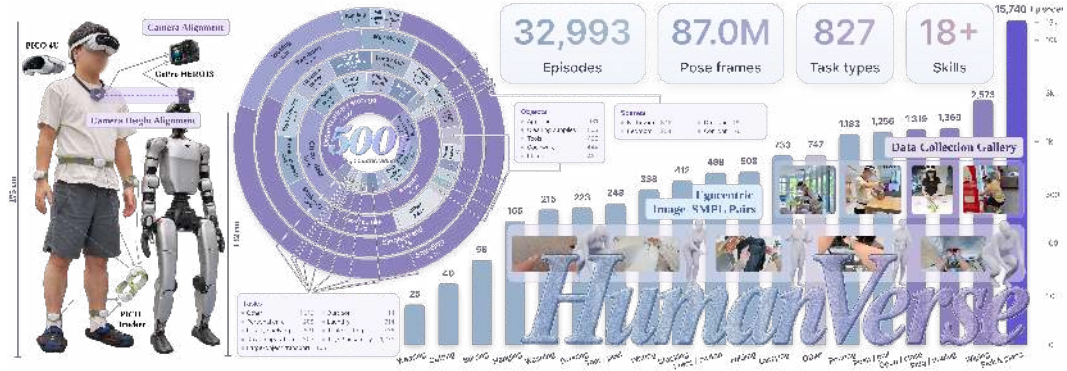


Figure 2: HUMANVERSE-500 overview: wearable capture setup, data composition, and synchronized egocentric image–SMPL pairs.

3 HUMANVERSE-500: EGOCENTRIC WHOLE-BODY HUMAN DATA

3.1 WHY SYNCHRONIZED WHOLE-BODY HUMAN DATA?

Hand-centric egocentric supervision captures interaction within reach but does not fully describe the approach, body repositioning, and transport needed for loco-manipulation. HUMANVERSE-500 couples first-person observations with whole-body motion so that these behaviors remain temporally aligned. Collection does not require the target humanoid, allowing human experience to scale independently of robot availability. The resulting data provide human motion supervision, not executable robot commands. Table 4 compares capture setups, motion coverage, and documented humanoid-training interfaces.

3.2 COLLECTION AND DATA COMPOSITION

HUMANVERSE-500 contains 500 hours of self-collected activity across varied scenes, objects, and mobile-manipulation behaviors. Collection prioritizes behavioral coverage beyond the downstream tasks. As shown in Fig. 2, a portable system records chest-mounted GoPro RGB and timestamped 24-joint PICO body kinematics. Dyn-HaMR reconstructs 21 hand joints from RGB using ViPE camera motion. We distinguish device-tracked body measurements from model-derived hand geometry; neither is treated as native robot action supervision. The corresponding training representations are defined in Sec. 4.2.

3.3 DATA SEPARATION AND EVALUATION SPLITS

A separate set contains 600 task-aligned human demonstrations, with 150 each for bottle disposal, single-hand toy storage, bimanual parts-box transport, and chair placement. These use the same capture protocol but enter only Stage III; they are excluded from the 500-hour corpus and its scaling subsets. Training and validation are partitioned by recording session before temporal window extraction. Scaling uses nested subsets of the frozen HUMANVERSE-500 training partition, while task-aligned human data and robot post-training data remain fixed. Robot evaluation configurations are excluded from human-data selection. Appendix B details the manifests, annotation provenance, coordinate conventions, and quality checks.

4 λ_0 : LEARNING HUMANOID CONTROL FROM HUMAN EXPERIENCE

We introduce λ_0 , a unified vision–language–action policy that transfers scalable human experience to humanoid control. Human data and robot trajectories provide complementary supervision but use incompatible state–action spaces. λ_0 resolves this mismatch by sharing a backbone and action expert while retaining domain-specific interfaces. Its three-stage training recipe learns interaction

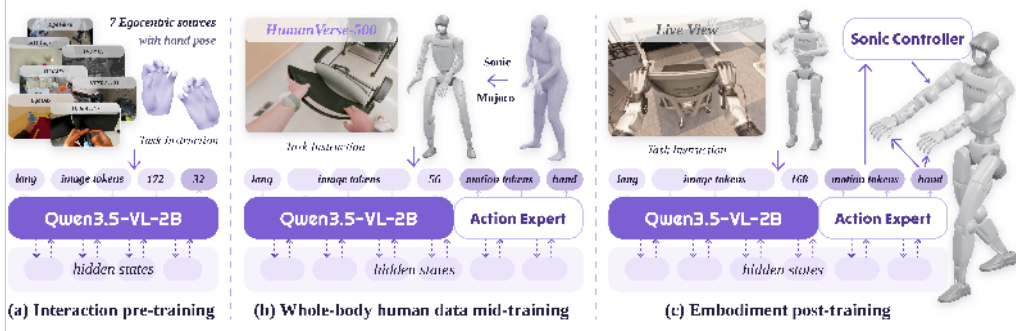


Figure 3: Three-stage human-to-humanoid transfer in λ_0 . Stage I learns interaction from egocentric data, Stage II couples body and hand motion with HUMANVERSE-500, and Stage III grounds the shared representation using task-aligned human and native robot trajectories. Each domain retains its own state–action interface.

structure, whole-body coordination, and executable robot control in sequence (Fig. 3). Human motion therefore transfers through shared representations rather than serving as actuator commands.

4.1 TRAINING DATA AND TRANSFER SETUP

We use four complementary data sources. Let $\mathcal{D}_E$ denote the union of public egocentric hand–object datasets, $\mathcal{D}_W$ denote the general whole-body corpus HUMANVERSE-500, $\mathcal{D}_T$ denote 600 separate task-aligned human demonstrations, and $\mathcal{D}_R^e$ denote native demonstrations for robot embodiment e . They provide interaction coverage, body–hand coordination, task alignment, and executable robot transitions while retaining native state–action semantics.

4.2 SHARED BACKBONE AND ACTION EXPERT

Shared transfer core. λ_0 consists of a Qwen3.5 vision–language backbone V_ϕ and a generative action expert F_ψ . The current state is quantized into 256 bins and appended to the language instruction. The backbone encodes this state-aware instruction together with the image $\mathbf{x}$, then supplies layer-wise features to the action expert. For each action space q , an input encoder E_q maps the noisy action and noise level into action tokens, while an output decoder D_q maps the shared prediction back to that action space:

$$\hat{\mathbf{a}}^q = D_q(F_\psi(V_\phi(\mathbf{x}, [\ell; Q(\mathbf{s}^q)]), E_q(\tilde{\mathbf{a}}^q, \sigma), e)). \quad (1)$$

Here e is the embodiment identifier. Sharing (V_ϕ, F_ψ) makes experience transferable across training stages. Keeping (E_q, D_q) separate prevents incompatible human kinematics and robot commands from being collapsed into a common coordinate system.

Interface-specific action semantics. The egocentric interface predicts relative wrist motion and wrist-local finger geometry. The whole-body human interface predicts body-motion tokens together with finger geometry. The robot-compatible interface instead pairs the same type of body-motion token with robot hand commands. These interfaces allow Stage II to learn coordinated body–hand dynamics rather than simply adding more hand-centric examples. Exact state–action definitions and layer-wise conditioning are provided in Appendices C.5 and C.6.

Masked action denoising. For $\epsilon \sim \mathcal{N}(0, I)$ and $\sigma \sim \text{Beta}(1.5, 1)$, we corrupt the action at horizon h as $\tilde{\mathbf{a}}_{h,\sigma} = (1 - \sigma_h)\mathbf{a}_h + \sigma_h\epsilon_h$ and train the expert to recover the clean action chunk. Let M^q contain all valid, non-prefix time–feature indices for action space q . The loss is the mean squared error over these scalar targets:

$$\mathcal{L}^q = \mathbb{E} \left[\frac{1}{|M^q|} \sum_{(h,j) \in M^q} \left(\hat{a}_{h,j}^q - a_{h,j}^q \right)^2 \right]. \quad (2)$$

The mask removes unavailable action dimensions, missing-hand labels, and clean RTC prefixes from supervision. Named body and hand blocks are reported as diagnostic metrics, but do not reweight the objective. Appendix C.2 gives the exact batch-level formulation.

4.3 THREE-STAGE TRAINING OBJECTIVES

Let $\theta^{(k)}$ denote the EMA checkpoint after Stage k . Each stage changes the supervision and active interface while initializing from the preceding checkpoint.

Stage I: learning interaction structure. We initialize the backbone from a vision–language checkpoint and train it jointly with the action expert and egocentric hand interface. Training uses EgoDex, EgoVerse, HO-Cap, HOI4D, HoloAssist, HOT3D, and VITRA-1M (Hoque et al., 2025; Punamiya et al., 2026; Wang et al., 2025; Liu et al., 2022; Wang et al., 2023b; Banerjee et al., 2025; Li et al., 2026a). Square-root sampling over usable frame counts N_d maintains coverage of smaller sources:

$$p_1(d) = \frac{\sqrt{N_d}}{\sum_{d' \in S_E} \sqrt{N_{d'}}}, \quad \theta^{(1)} = \arg \min_{\theta} \mathbb{E}_{d \sim p_1, \tau \sim \mathcal{D}_d} [\mathcal{L}^E(\tau; \theta)]. \quad (3)$$

The targets are relative wrist motion and local finger geometry, which provide broad hand–object interaction supervision before whole-body training.

Stage II: coupling whole-body motion and interaction. For each HUMANVERSE-500 episode, offline transformations Φ_H and Φ_R produce paired human and robot-compatible views that share RGB, language, and timestamps. The human view combines body-motion tokens with finger geometry. For the robot-compatible view, a frozen SONIC motion encoder and G1 MuJoCo rollout produce motion tokens and robot states, while deterministic hand retargeting maps reconstructed MANO-21 geometry to Revo2 commands (Luo et al., 2025). We sample the human and robot-compatible views equally and optimize

$$\theta^{(2)} = \arg \min_{\theta \leftarrow \text{EMA}(\theta^{(1)})} \left(\frac{1}{2} \mathbb{E}_{\Phi_H(\mathcal{D}_W)} [\mathcal{L}^H] + \frac{1}{2} \mathbb{E}_{\Phi_R(\mathcal{D}_W)} [\mathcal{L}^R] \right). \quad (4)$$

The shared model and whole-body interfaces are updated while the Stage-I hand head is frozen. All supervision in this stage comes from human recordings, with no task-aligned or native robot demonstrations.

Stage III: grounding tasks and robot execution. We combine task-aligned human demonstrations, which preserve interaction coverage, with native robot trajectories, which provide executable state–action transitions. For human sampling coefficient α , embodiment grounding is formalized as

$$\theta_e^{(3)} = \arg \min_{\theta \leftarrow \text{EMA}(\theta^{(2)})} \left(\alpha \mathbb{E}_{\mathcal{D}_T} [\mathcal{L}^H] + (1 - \alpha) \mathbb{E}_{\mathcal{D}_R^e} [\mathcal{L}^R] \right). \quad (5)$$

The shared model, embodiment conditioning, and active interfaces are optimized jointly. The default 2B model is trained for 200k, 100k, and 50k updates across the three stages with a 50-step action horizon. During scaling experiments, the Stage-III mixture and update budget remain fixed unless robot data are the controlled variable. Appendix C.3 provides the complete optimization and sampling settings.

Deployment. At inference time, λ_0 takes the target robot’s observation, instruction, proprioceptive state, and embodiment identifier, and activates only its robot interface. For G1, SONIC interprets the predicted body-motion tokens and executes them together with the predicted hand commands. Human joints are never sent directly to the robot, and human observations are not required at deployment. The same shared policy can retain a separate native interface for each additional embodiment.

5 EXPERIMENTS

We evaluate whether the three-stage recipe turns human data into improved humanoid control and object transfer. Specifically, we ask: **Q1.** *How well does the model perform on humanoid locomotion?* (Sec. 5.2); **Q2.** *How do human-data scale and the training recipe affect learning?* (Sec. 5.3); **Q3.** *Does human-data scaling improve downstream transfer?* (Sec. 5.4). Accordingly, we evaluate held-out learning dynamics, task performance, and unseen-object generalization.

5.1 EVALUATION SETUP AND METRICS

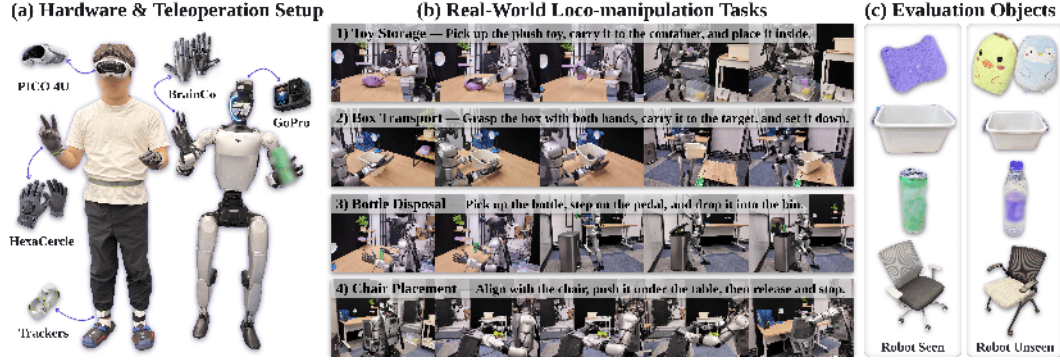


Figure 4: **Real-world data collection and evaluation.** (a) Egocentric teleoperation collects whole-body demonstrations. (b) Four loco-manipulation tasks evaluate task success. (c) Robot-seen and robot-unseen objects are used for standard task and object-generalization evaluation, respectively.

Hardware. As shown in Fig. 4(a), a PICO 4U with five trackers captures body motion, which SONIC converts into Unitree G1 control signals. Finger motion from the HexaCercle M11 gloves is mapped to control the BrainCo Revo 2 hands. A GoPro mounted on the G1’s head provides egocentric RGB, while the G1’s computer records synchronized whole-body expert demonstrations.

Evaluation. We evaluate six core SIMPLE tasks (Wei et al., 2026a) at Levels 0–2 with ten rollouts per task–level pair, and four real-world tasks shown in Fig. 4(b), with ten frozen initial configurations per task and policy. We report task success rate and task progress. Progress is computed as the sum of fixed milestone scores, normalized to 100 per task and averaged equally across tasks. Standard trials use robot-seen objects in Fig. 4(c). Generalization uses robot-unseen objects appearing only in human demonstrations. Appendix F details the tasks, initialization, and scoring.

Baselines. In simulation, we compare against the public SIMPLE leaderboard baselines (Wei et al., 2026a). Public entries retain their source protocols, whereas StarVLA and λ_0 are evaluated by us. In the real world, we compare against $\pi_{0.5}$ (Physical Intelligence et al., 2025), GR00T N1.6 (GEAR Team et al., 2025), Ψ_0 (Wei et al., 2026b), and StarVLA (StarVLA Community, 2026) under the same protocol. Appendix C provides training and implementation details.

5.2 Q1: HOW WELL DOES THE MODEL PERFORM ON HUMANOID LOCO-MANIPULATION?

Real-world evaluation. Table 1 reports all fully adjudicated real-world trials. Our λ_0 achieves the highest four-task average success rate (62.5%, 25/40 trials) and weighted progress (80.4%), exceeding the strongest external baselines by 22.5 and 25.8 percentage points, respectively. λ_0 records the highest success count on every task, with its largest success-rate margin on bottle disposal. Against the strongest external result per task, it improves progress by 51.0 points on bottle disposal, 16.5 on toy storage, and 13.5 on box transport. On chair placement, it completes 6/10 trials versus StarVLA’s 3/10 despite lower progress (77.0% versus 82.5%), separating terminal completion from partial progress.

Table 1: **Real-world task performance.** We report success rates over ten trials and weighted progress (%) for each task. Averages weight the four tasks equally.

Method	Bottle Disposal		Toy Storage		Box Transport		Chair Placement		Average	
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	2/10	28.0	5/10	64.5	5/10	58.0	4/10	65.5	16/40	54.0
GR00T N1.6 (GEAR Team et al., 2025)	0/10	10.0	1/10	19.0	0/10	10.0	2/10	46.0	3/40	21.3
Ψ_0 (Wei et al., 2026b)	1/10	26.5	2/10	30.0	1/10	37.5	0/10	33.0	4/40	31.8
StarVLA (StarVLA Community, 2026)	0/10	10.0	5/10	55.0	3/10	71.0	3/10	82.5	11/40	54.6
λ_0 (Ours)	5/10	79.0	7/10	81.0	7/10	84.5	6/10	77.0	25/40	80.4

Table 2: **SIMPLE task performance.** Success rates (%) at Levels 0–2 with ten trials per task–level pair. Average is computed per level. Overall averages all 18 task–level pairs.

Model	XMovePick	BendPick	Handover	Mobile P&P	Grasp	XMoveBendPick	Average	Overall
H-RDT (Bi et al., 2025)	0/0/20	0/0/10	0/10/0	0/0/0	0/0/0	0/0/0	0.0/1.7/5.0	2.2
InternVLA-M1 (Chen et al., 2025)	0/0/0	50/50/0	0/0/0	0/0/0	0/0/0	30/50/70	13.3/16.7/11.7	13.9
EgoVLA (Yang et al., 2025)	0/10/20	70/50/80	0/40/30	0/0/0	100/100/70	30/50/40	33.3/41.7/40.0	38.3
Diffusion Policy (Chi et al., 2025)	30/30/20	100/80/60	30/20/40	40/0/0	80/90/80	0/0/0	46.7/36.7/33.3	38.9
GR00T N1.6 (GEAR Team et al., 2025)	100/100/70	70/70/60	10/30/30	0/0/0	90/90/70	40/40/10	51.7/55.0/40.0	48.9
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	70/50/10	100/100/80	50/40/50	30/30/30	100/100/80	0/0/0	58.3/53.3/41.7	51.1
DreamZero (Ye et al., 2026)	100/100/100	90/90/80	70/80/90	50/30/30	90/100/70	0/0/10	66.7/66.7/63.3	65.6
ACT (Zhao et al., 2023)	100/100/50	100/90/90	70/70/100	50/50/50	100/100/80	90/100/100	85.0/85.0/78.3	82.8
Ψ_0 (Wei et al., 2026b)	100/100/60	100/100/100	70/70/100	70/50/60	100/100/80	100/90/90	90.0/85.0/81.7	85.6
StarVLA (StarVLA Community, 2026)	100/90/90	70/90/70	80/80/100	90/80/80	90/100/70	70/100/100	83.3/90.0/85.0	86.1
λ_0 (Ours)	100/100/80	100/100/100	90/100/100	90/70/70	100/90/80	60/100/90	90.0/93.3/86.7	90.0

Simulation evaluation. Table 2 reports 90.0% overall success for λ_0 , with 90.0%, 93.3%, and 86.7% at Levels 0, 1, and 2. Its overall score exceeds StarVLA by 3.9 percentage points and Ψ_0 by 4.4 points. λ_0 leads StarVLA at all three levels; it matches Ψ_0 at Level 0 and leads at Levels 1 and 2. It also exceeds ACT at every level. These results use the best recorded configuration for each task, as detailed in Appendix E.

5.3 Q2: HOW DO HUMAN-DATA SCALE AND THE TRAINING RECIPE AFFECT LEARNING?

Validation-loss scaling. To isolate human-data scale in Stage II, we train the same 2B model for 100k updates on nested 5%, 10%, 25%, 50%, and 100% subsets with a fixed seed and optimization schedule. Every 1k updates, we evaluate two fixed, session-disjoint holdouts: *clean* contains sessions with upper-quartile whole-body and bilateral-hand annotation validity, while *representative* is stratified by data source and annotation validity to reflect the full data distribution. Following EgoScale (Zheng et al., 2026b), we analyze training dynamics and minimum loss at each scale.

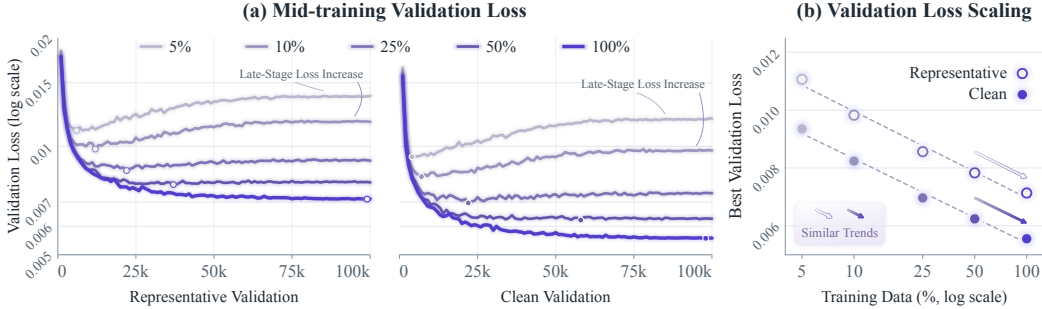


Figure 5: **Human-data scaling of validation loss.** (a) Training curves on representative and clean holdouts at five data fractions. (b) Minimum validation loss versus data scale with descriptive log-linear fits over the measured range. Minima are selected independently on each holdout.

Training dynamics. Fig. 5(a) shows a clear scale-dependent change in the useful training horizon. On the representative holdout, the 5% and 10% runs reach their minima at 6k and 12k updates, but finish 24.7% and 19.0% above those minima. The clean holdout shows the same behavior, with final losses 27.6% and 18.1% above the respective minima. In contrast, the 50% and 100% runs finish within 1.3% of their best representative loss and within 0.9% on clean. Their minima also move toward the end of training. Thus, additional human data both lowers validation error and delays the late-stage degradation observed at smaller scales. The 25% run forms an intermediate regime: both holdouts reach their minima near 22k updates and finish only 6.4–6.5% higher. Comparing only final checkpoints would therefore confound data scale with overtraining.

Scaling trend. Fig. 5(b) summarizes the best checkpoint at each scale. From 5% to 100%, minimum loss falls by 35.5% on representative and 40.6% on clean. Fitting $\hat{L}_h(D) = a_h + b_h \ln D$ gives $R^2 = 0.9868/0.9901$ and per-doubling reductions of 0.000900/0.000877 for representative/clean. Clean remains lower at every scale, but the fitted slope magnitudes differ by just 2.7%, suggesting that annotation quality shifts the attainable loss more than the response to additional data. With one run per scale and downstream results pending, these fits do not establish cross-seed reproducibility, extrapolation, or improved robot control. Appendix D.1 gives the full protocol. The corresponding downstream evaluation is outlined in Sec. 5.4 and Fig. 6.

Table 3: **Training-stage ablations with fixed robot supervision.** Filled/open circles indicate included/removed data; metrics report SIMPLE success and real-world averages across four tasks. The Full Recipe SIMPLE row uses the task-wise best configurations described in Appendix E.

Variant	S1	S2	Robot	SIMPLE (%)				Real-world (%)	
				L0	L1	L2	Overall	Success	Progress
w/o human-data pretraining	○	○	●	83.3	83.3	71.7	79.4	15.0	54.0
w/o Stage II mid-training	●	○	●	83.6	86.7	76.7	82.3	22.5	59.9
w/o Stage I pretraining	○	●	●	88.0	91.7	83.7	87.8	50.0	75.9
λ_0 (Full Recipe)	●	●	●	90.0	93.3	86.7	90.0	62.5	80.4

Training-stage ablations. Table 3 compares variants with and without the two human-data stages. SIMPLE success is 90.0% with the full recipe, 87.8% without Stage I, 82.3% without Stage II, and 79.4% without both. The full-recipe SIMPLE entry uses the task-wise best configurations described in Appendix E. In the matched real-world evaluation, success/progress is 15.0%/54.0% without human-data pretraining, 50.0%/75.9% without Stage I, and 22.5%/59.9% without Stage II, compared with 62.5%/80.4% for the full recipe. Removing Stage II causes a larger success-rate drop than removing Stage I (40.0 versus 12.5 percentage points). All variants retain the vision–language initialization and Stage III. Appendix H details controls and failure modes.

5.4 Q3: DOES HUMAN-DATA SCALING IMPROVE DOWNSTREAM TRANSFER?

Downstream task-progress scaling. Fig. 6 anchors the scaling comparison with the no-Stage-II and full-recipe results from Table 3: real-world task progress rises from 59.9% to 80.4%. Evaluations at 5%, 10%, 25%, and 50% are pending. The comparison fixes Stage-I initialization, model capacity, downstream demonstrations, Stage-III mixture and update budget, and evaluation configurations. Nonzero fractions share the Stage-II update budget and checkpoint-selection rule. The two endpoints establish a 20.5-point gain from Stage II, but do not yet establish a monotonic trend across intermediate data scales.

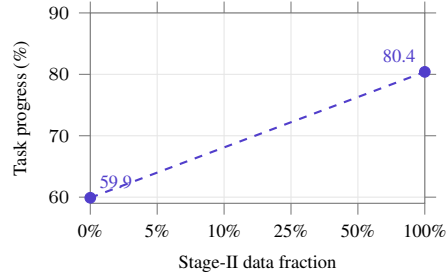


Figure 6: Real-world progress vs. Stage-II data.

Object generalization. We further evaluate object generalization on real-world tasks. One policy is trained on robot A and human A/B demonstrations. *Seen* (A + A + B) tests A, and *Human-guided* (A + A + B) tests B. *Zero-shot* (A + B) trains on robot A only and tests B with matched initializations and scoring, measuring transfer from human data to robot-unseen objects (Appendix I).

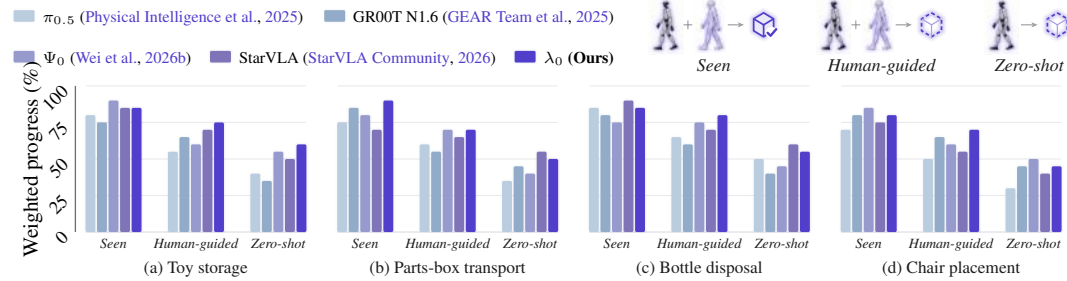


Figure 7: Object-transfer progress. Seen and Human-guided test one joint policy on robot-seen A and human-only B. Zero-shot tests a robot-only policy on B.

Cross-scale evaluation controls object membership to separate exposure from scale. Appendix Sections H–J provide controls, safety statistics, and failure analysis.

6 CONCLUSION

We presented λ_0 , a three-stage recipe for transferring egocentric whole-body human experience to humanoid loco-manipulation. HUMANVERSE-500 provides 500 hours of synchronized visual and kinematic data, while the staged training protocol separates human interaction priors, whole-body motion, and humanoid control. Our evaluation spans validation-loss scaling, task success, and cross-object transfer in simulation and on four real-world tasks. The study is limited to the evaluated embodiment, task families, and collection distribution. Reconstructed hand motion, imperfect human-to-robot dynamics, and finite object coverage remain directions for future work.

AI USE STATEMENT

Generative AI tools were used to assist with experimental code development and debugging, data analysis, manuscript drafting and language editing, and figure preparation. All reported experimental results were obtained through actual model training and evaluation, not generated by AI tools. The authors reviewed and verified all AI-assisted content and take full responsibility for the methods, results, and conclusions.

REFERENCES

- Joao Pedro Araujo, Yanjie Ze, Pei Xu, Jiajun Wu, and C. Karen Liu. Retargeting matters: General motion retargeting for humanoid motion tracking. *arXiv preprint arXiv:2510.02252*, 2025.
- Shikhar Bahl, Abhinav Gupta, and Deepak Pathak. Human-to-robot imitation in the wild. In *Proceedings of Robotics: Science and Systems*, 2022. doi: 10.15607/RSS.2022.XVIII.026.
- Prithviraj Banerjee et al. HOT3D: Hand and object tracking in 3d from egocentric multi-view videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7061–7071, 2025.
- Homanga Bharadhwaj, Abhinav Gupta, Shubham Tulsiani, and Vikash Kumar. Zero-shot robot manipulation from passive human videos. *arXiv preprint arXiv:2302.02011*, 2023.
- Hongzhe Bi, Lingxuan Wu, Tianwei Lin, Hengkai Tan, Zhizhong Su, Hang Su, and Jun Zhu. H-RDT: Human manipulation enhanced bimanual robotic manipulation. *arXiv preprint arXiv:2507.23523*, 2025.
- Anthony Brohan et al. RT-1: Robotics transformer for real-world control at scale. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.025.
- Qingwen Bu et al. AgiBot World Colosseo: A large-scale manipulation platform for scalable and intelligent embodied systems. *arXiv preprint arXiv:2503.06669*, 2025.
- Xinyi Chen, Yilun Chen, Yanwei Fu, Ning Gao, Jiaya Jia, Weiyang Jin, Hao Li, Yao Mu, Jiangmiao Pang, Yu Qiao, et al. InternVLA-M1: A spatially guided vision-language-action framework for generalist robot policy. *arXiv preprint arXiv:2510.13778*, 2025.
- Cheng Chi, Zhenjia Xu, Chuer Pan, Eric Cousineau, Benjamin Burchfiel, Siyuan Feng, Russ Tedrake, and Shuran Song. Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots. In *Proceedings of Robotics: Science and Systems*, 2024.
- Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2025. doi: 10.1177/02783649241273668.
- Dima Damen et al. Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100. *International Journal of Computer Vision*, 130:33–55, 2022. doi: 10.1007/s11263-021-01531-2.
- Sudeep Dasari, Frederik Ebert, Stephen Tian, Suraj Nair, Bernadette Bucher, Karl Schmeckpeper, Siddharth Singh, Sergey Levine, and Chelsea Finn. RoboNet: Large-scale multi-robot learning. In *Proceedings of the Conference on Robot Learning*, pp. 885–897, 2020.
- Shivin Dass, Karl Pertsch, Hejia Zhang, Youngwoon Lee, Joseph J. Lim, and Stefanos Nikolaidis. PATO: Policy assisted teleoperation for scalable robot data collection. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.013.
- Daniel DeTone, Federica Bogo, Eric-Tuan Le, Duncan Frost, Julian Straub, Yawar Siddiqui, Yuting Ye, Jakob Engel, Richard Newcombe, and Lingni Ma. NymeriaPlus: Enriching nymeria dataset with additional annotations and data. *arXiv preprint arXiv:2603.18496*, 2026. URL <https://arxiv.org/abs/2603.18496>.

- Figure. Introducing Index: Building the world’s largest and most diverse physical dataset. <https://www.figure.ai/news/introducing-index>, August 2026. Accessed September 18, 2026.
- Figure AI. Introducing helix 02: Full-body autonomy. <https://www.figure.ai/news/helix-02>, January 2026.
- Zipeng Fu, Qingqing Zhao, Qi Wu, Gordon Wetzstein, and Chelsea Finn. HumanPlus: Humanoid shadowing and imitation from humans. In *Proceedings of the 8th Conference on Robot Learning*, pp. 2828–2844, 2025a.
- Zipeng Fu, Tony Z. Zhao, and Chelsea Finn. Mobile ALOHA: Learning bimanual mobile manipulation using low-cost whole-body teleoperation. In *Proceedings of the 8th Conference on Robot Learning*, pp. 4066–4083, 2025b.
- GEAR Team, Allison Azzolini, Johan Bjorck, et al. GR00T N1.6: An improved open foundation model for generalist humanoid robots. NVIDIA Research, 2025. URL https://research.nvidia.com/labs/gear/gr00t-n1_6/.
- Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Quan Vuong, Ted Xiao, Pannag R. Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An open-source generalist robot policy. In *Proceedings of Robotics: Science and Systems*, 2024.
- Raghav Goyal et al. The “something something” video database for learning and evaluating visual common sense. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 5842–5850, 2017.
- Kristen Grauman et al. Ego4D: Around the world in 3,000 hours of egocentric video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Kristen Grauman et al. Ego-Exo4D: Understanding skilled human activity from first- and third-person perspectives. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 19383–19400, 2024.
- Tairan He, Zhengyi Luo, Xialin He, Wenli Xiao, Chong Zhang, Weinan Zhang, Kris M. Kitani, Changliu Liu, and Guanya Shi. OmniH2O: Universal and dexterous human-to-humanoid whole-body teleoperation and learning. In *Proceedings of the 8th Conference on Robot Learning*, pp. 1516–1540, 2025.
- Ryan Hoque, Peide Huang, David J. Yoon, Mouli Sivapurapu, and Jian Zhang. EgoDex: Learning dexterous manipulation from large-scale egocentric video. *arXiv preprint arXiv:2505.11709*, 2025.
- Yingdong Hu et al. OpenHLM: An empirical recipe for whole-body humanoid loco-manipulation. *arXiv preprint arXiv:2606.22174*, 2026.
- Yifei Huang et al. EgoExoLearn: A dataset for bridging asynchronous ego- and exo-centric view of procedural activities in real world. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22072–22086, 2024.
- Eric Jang, Alex Irpan, Mohi Khansari, Daniel Kappler, Frederik Ebert, Corey Lynch, Sergey Levine, and Chelsea Finn. BC-Z: Zero-shot task generalization with robotic imitation learning. In *Proceedings of the 5th Conference on Robot Learning*, pp. 991–1002, 2022.
- Haoran Jiang, Jin Chen, Qingwen Bu, Li Chen, Modi Shi, Yanjie Zhang, Delong Li, Chuanzhe Suo, Chuang Wang, Zhihui Peng, and Hongyang Li. WholeBodyVLA: Towards unified latent VLA for whole-body loco-manipulation control. In *International Conference on Learning Representations*, 2026.
- Simar Kareer, Dhruv Patel, Ryan Punamiya, Pranay Mathur, Shuo Cheng, Chen Wang, Judy Hoffman, and Danfei Xu. EgoMimic: Scaling imitation learning via egocentric video. *arXiv preprint arXiv:2410.24221*, 2024.

- Alexander Khazatsky et al. DROID: A large-scale in-the-wild robot manipulation dataset. In *Proceedings of Robotics: Science and Systems*, 2024.
- Moo Jin Kim et al. OpenVLA: An open-source vision-language-action model. In *Proceedings of the 8th Conference on Robot Learning*, pp. 2679–2713, 2025.
- Ashish Kumar, Saurabh Gupta, and Jitendra Malik. Learning navigation subroutines from egocentric videos. In *Proceedings of the Conference on Robot Learning*, pp. 617–626, 2020.
- Marion Lepert, Jiaying Fang, and Jeannette Bohg. Phantom: Training robots without robots using only human videos. In *Proceedings of the 9th Conference on Robot Learning*, pp. 4545–4565, 2025.
- Qixiu Li et al. Scalable vision-language-action model pretraining for robotic manipulation with real-life human activity videos. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2026a.
- Zhe Li, Zhenzhe Zhang, Yangyang Wei, Wenjie Zhang, Xichen Yuan, Peiyuan Zhi, Gen Li, Xinying Guo, Fengjie Gao, Jianfei Yang, and Shanghang Zhang. ω -0: A latent predictive world action model for concurrent humanoid loco-manipulation. *arXiv preprint arXiv:2608.06375*, 2026b.
- Lightwheel AI. EgoSuite-Open100K: 100,000 hours of egocentric human data for physical AI. <https://huggingface.co/blog/LightwheelAI/egosuite-open100k>, August 2026. August 26 release announcement; accessed September 18, 2026.
- Yunze Liu, Yun Liu, Che Jiang, Kangbo Lyu, Weikang Wan, Hao Shen, Boqiang Liang, Zhoujie Fu, He Wang, and Li Yi. HOI4D: A 4d egocentric dataset for category-level human-object interaction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 21013–21022, 2022.
- Zhengyi Luo, Jinkun Cao, Alexander Winkler, Kris Kitani, and Weipeng Xu. Perpetual humanoid control for real-time simulated avatars. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 10895–10904, 2023.
- Zhengyi Luo et al. SONIC: Supersizing motion tracking for natural humanoid whole-body control. *arXiv preprint arXiv:2511.07820*, 2025.
- Ajay Mandlekar, Danfei Xu, Josiah Wong, Soroush Nasiriany, Chen Wang, Rohun Kulkarni, Li Fei-Fei, Silvio Savarese, Yuke Zhu, and Roberto Martín-Martín. What matters in learning from offline human demonstrations for robot manipulation. In *Proceedings of the 5th Conference on Robot Learning*, pp. 1678–1690, 2022.
- Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3M: A universal visual representation for robot manipulation. In *Proceedings of the 6th Conference on Robot Learning*, pp. 892–909, 2023.
- Open X-Embodiment Collaboration. Open X-embodiment: Robotic learning datasets and RT-X models. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2024. doi: 10.1109/ICRA57147.2024.10611477.
- Carolina Parada. Gemini Robotics 2 brings whole body intelligence to robots. <https://deepmind.google/blog/gemini-robotics-2-brings-whole-body-intelligence-to-robots/>, July 2026.
- Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, et al. $\pi_{0.5}$: A vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.
- Ryan Punamiya et al. EgoVerse: An egocentric human dataset for robot learning from around the world. *arXiv preprint arXiv:2604.07607*, 2026.

- Yuzhe Qin, Wei Yang, Binghao Huang, Karl Van Wyk, Hao Su, Xiaolong Wang, Yu-Wei Chao, and Dieter Fox. AnyTeleop: A general vision-based dexterous robot arm-hand teleoperation system. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.015.
- Francesco Ragusa, Antonino Furnari, Salvatore Livatino, and Giovanni Maria Farinella. The MEC-CANO dataset: Understanding human-object interactions from egocentric videos in an industrial-like domain. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 1569–1578, 2021.
- Fadime Sener, Dibyadip Chatterjee, Daniel Shelepov, Kun He, Dipika Singhania, Robert Wang, and Angela Yao. Assembly101: A large-scale multi-view video dataset for understanding procedural activities. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 21096–21106, 2022.
- Lin Shao, Toki Migimatsu, Qiang Zhang, Kaiyuan Yang, and Jeannette Bohg. Concept2Robot: Learning manipulation concepts from instructions and human demonstrations. In *Proceedings of Robotics: Science and Systems*, 2020. doi: 10.15607/RSS.2020.XVI.082.
- Modi Shi et al. EgoHumanoid: Unlocking in-the-wild loco-manipulation with robot-free egocentric demonstration. *arXiv preprint arXiv:2602.10106*, 2026.
- Sumedh Sontakke, Jesse Zhang, Séb Arnold, Karl Pertsch, Erdem Bıyık, Dorsa Sadigh, Chelsea Finn, and Laurent Itti. RoboCLIP: One demonstration is enough to learn robot policies. In *Advances in Neural Information Processing Systems*, 2023.
- StarVLA Community. StarVLA: A lego-like codebase for vision-language-action model developing. *arXiv preprint arXiv:2604.05014*, 2026.
- Symbiosis Robotics. DPC. <https://symbiosis-robotics.com/research/dpc/en/>, 2026. Project page, accessed August 19, 2026.
- Homer Rich Walke, Kevin Black, Tony Z. Zhao, Quan Vuong, Chongyi Zheng, Philippe Hansen-Estruch, Andre Wang He, Vivek Myers, Moo Jin Kim, Max Du, Abraham Lee, Kuan Fang, Chelsea Finn, and Sergey Levine. BridgeData V2: A dataset for robot learning at scale. In *Proceedings of the 7th Conference on Robot Learning*, pp. 1723–1736, 2023.
- Chen Wang, Linxi Fan, Jiankai Sun, Ruohan Zhang, Li Fei-Fei, Danfei Xu, Yuke Zhu, and Anima Anandkumar. MimicPlay: Long-horizon imitation learning by watching human play. In *Proceedings of the 7th Conference on Robot Learning*, pp. 201–221, 2023a.
- Jikai Wang, Qifan Zhang, Yu-Wei Chao, Bowen Wen, Xiaohu Guo, and Yu Xiang. HO-cap: A capture system and dataset for 3d reconstruction and pose tracking of hand-object interaction. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2025.
- Xin Wang et al. HoloAssist: An egocentric human interaction dataset for interactive AI assistants in the real world. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023b.
- Ye Wang, Pei Lin, Xiong-Hui Chen, et al. Ego2Robot: Scalable robot data synthesis from egocentric human data. *arXiv preprint arXiv:2608.02580*, 2026.
- Songlin Wei, Zhenhao Ni, Jie Liu, Zhenyu Zhao, Junjie Ye, Hongyi Jing, Junkai Xia, Xiawei Liu, Michael Leong, Liang Heng, Di Huang, and Yue Wang. SIMPLE: Simulation-based policy learning and evaluation for humanoid loco-manipulation. *arXiv preprint arXiv:2606.08278*, 2026a.
- Songlin Wei et al. ψ_0 : An open foundation model towards universal humanoid loco-manipulation. *arXiv preprint arXiv:2603.12263*, 2026b.
- Xiaomi Robotics Team, Jun Guo, Piaopiao Jin, et al. Xiaomi-Robotics-1: Scaling vision-language-action models with over 100k hours of real-world trajectories. *arXiv preprint arXiv:2607.15330*, 2026.

- Ruihan Yang, Qinxu Yu, Yecheng Wu, Rui Yan, Borui Li, An-Chieh Cheng, Xueyan Zou, Yunhao Fang, Xuxin Cheng, Ri-Zhao Qiu, et al. EgoVLA: Learning vision-language-action models from egocentric human videos. *arXiv preprint arXiv:2507.12440*, 2025.
- Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026.
- Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.016.
- Jia Zheng, Teli Ma, Yudong Fan, Zifan Wang, Shuo Yang, and Junwei Liang. MotionWAM: Towards foundation world action models for real-time humanoid loco-manipulation. *arXiv preprint arXiv:2606.09215*, 2026a. URL <https://arxiv.org/abs/2606.09215>.
- Ruijie Zheng et al. EgoScale: Scaling dexterous manipulation with diverse egocentric human data. *arXiv preprint arXiv:2602.16710*, 2026b.
- Brianna Zitkovich et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of the 7th Conference on Robot Learning*, pp. 2165–2183, 2023.

A SUPPLEMENTARY MATERIALS

B DATASET SCOPE, MANIFEST, AND PROVENANCE

The three training stages use disjointly defined data resources. Stage I draws on the seven heterogeneous egocentric datasets listed in the main paper; these public sources are not counted as part of HUMANVERSE-500. HUMANVERSE-500 denotes only the 500-hour synchronized egocentric whole-body corpus used in Stage II. The 600 demonstrations collected for the four downstream tasks form a third resource used only in Stage III and are excluded from both the HUMANVERSE-500 duration and its scaling subsets. This separation prevents broad human experience, task-aligned human supervision, and robot grounding from being conflated in the scaling analysis.

The frozen HUMANVERSE-500 manifest records each episode’s participant, recording session, environment, activity label, duration, video frame count, body-tracking coverage, per-hand visibility, coordinate metadata, conversion version, and train or validation assignment. Temporal windows inherit the split of their source episode and are never partitioned independently. A separate manifest records the 600 task-aligned demonstrations and their downstream task identities.

An episode is structurally valid only when timestamps are monotonic, video and tracking intervals overlap, coordinate conventions are declared, and every unmasked value is finite. Missing or occluded hands mask only the affected supervision blocks. Reprojection error, estimator confidence, body-tracking coverage, and conversion failures remain explicit diagnostic fields rather than undocumented sampling filters.

B.1 ANNOTATION PROVENANCE AND COORDINATE RECORDS

The wearable body stream is synchronized device kinematics, not optical motion-capture ground truth. Dyn-HaMR reconstructs 21 joints per hand from RGB using ViPE camera motion. Records retain the camera trajectory used for estimation, per-track visibility, and diagnostic confidence. Raw streams are the canonical dataset record; derived views are versioned with source-frame indices and coordinate transforms. Unavailable or occluded observations remain masked rather than being replaced by temporally imputed poses.

Camera-to-world and body-frame metadata are preserved whenever available. An episode-fixed hand-trajectory view is anchored at the first valid camera pose to retain global displacement, while current-camera coordinates support geometric verification. These stored views are distinct from the query-time camera frame and relative wrist targets used by the training interface in Sec. 4.2. Finger geometry is wrist-local, palm-aligned, and scale-normalized. Whole-body tokens are computed from the tracked temporal sequence using a frozen motion encoder, not from an episode-level pose summary. Body and hand targets retain separate validity masks because their sensors and estimators have different failure modes.

B.2 QUALITY ASSURANCE AND SPLIT CONSTRUCTION

Structural checks verify stream lengths, timestamp monotonicity, finite values, coordinate conventions, and source identity. Geometric checks reproject reconstructed hands into RGB and record reprojection and visibility diagnostics. A synchronized viewer exposes RGB, current-camera 3D, episode-frame 3D, and body trajectories for manual audit. Diagnostics affect inclusion only through the pre-specified data contract: invalid target dimensions are masked and conversion failures are reported explicitly.

The 600 task-aligned demonstrations have an independent session-level partition. Separate session, object-instance, and evaluation-configuration identifiers support experimental separation; the frozen robot evaluation manifest is not consulted during human-data selection. Dataset reporting includes duration, episode count, participant and environment counts, activity distribution, bilateral-hand visibility, body-tracking coverage, and conversion failure rate. Nested scaling subsets are drawn only from the frozen general-purpose training partition, as detailed in Appendix D.

Table 4: Egocentric human-data resources and documented training interfaces. Body and hand coverage refers to supervision, not what appears in RGB. NR: not reported in the cited source; this does not imply incompatibility.

Resource	Hours	Body / hands	SMPL	Humanoid interface	Capture setup
EgoDex (Hoque et al., 2025)	829	Upper body; articulated hands	NR	NR in dataset release	Apple Vision Pro
HOT3D (Banerjee et al., 2025)	13.9	Hands and objects; no full body	NR	NR	Aria / Quest 3; external tracking
NymeriaPlus (DeTone et al., 2026)	NR ^a	Full body; finger motion NR	Provided	NR	Aria, wristbands, XSens
EgoHumanoid (Shi et al., 2026)	NR	Full body; articulated hands	NR	G1: wrist targets + locomotion commands	PICO + trackers; ZED X Mini
EgoScale (Zheng et al., 2026b)	20,854	Wrist and hand action supervision	NR	G1 transfer; wrist / hand interface	Multiple human-data sources
Index (Figure, 2026)	NR	Pose schema NR	NR	Figure / Helix; G1 NR	Distributed creator recordings
EgoSuite-Open100K (Lightwheel AI, 2026)	10k ^b	Hands; full body in body subsets	NR	NR; LeRobot is a storage format	Head camera; wrist views in Pro
HUMANVERSE-500 (Ours)	500	Tracked full body; reconstructed hands	Compatible	SONIC body tokens; G1 training views	Portable PICO + GoPro

^aNymeriaPlus upgrades the 300+ h Nymeria corpus; its retained duration after sequence filtering is not separately specified here.

^bThe August 26, 2026 announcement reports 10k h initially available out of 100k h planned, including 12k h of planned full-body subsets; these planned hours are not counted as released full-body data. SMPL compatibility and a demonstrated robot interface are distinct. Our body stream is device-derived kinematics, not optical motion-capture ground truth; hand geometry is reconstructed offline. Capture hardware is listed without an unverified cost ranking.

Table 5: Problem-level comparison with the closest humanoid VLA studies. “General” denotes human data not collected for the downstream evaluation tasks; “task-targeted” denotes demonstrations whose task or object coverage is selected for the downstream study. This table compares experimental questions rather than reported scores.

Method	Human signal	Role in training	Controlled scale	Primary question
WholeBodyVLA (Jiang et al., 2026)	300 h action-free egocentric video	latent locomotion pre-training; robot manipulation latents and teleop grounding	0–100% video × teleop trajectories	Can visual latent actions reduce locomanipulation teleoperation?
EgoHumanoid (Shi et al., 2026)	same-task RGB and wearable motion	direct human–robot co-training after view/action alignment	0–300 human demonstrations and sampling ratio	Can task-matched human data improve scene generalization?
Ψ_0 (Wei et al., 2026b)	829 h wrist/fingertip-labeled EgoDex video	task-space VLM pre-training before robot joint-action learning	full EgoDex versus 10%	Which staged data recipe yields an effective humanoid foundation model?
OpenHLM (Hu et al., 2026)	task-targeted stationary HuMI motion	heterogeneous co-training after whole-body VLA adaptation	whole-body teleop count and data-source choice	Which interfaces and data sources yield a whole-body-native VLA?
λ_0	500 h synchronized RGB, full-body motion, and bilateral hands	general mid-training, separated from task-aligned human and robot post-training	0–100% human data × nested robot budget	What scales with explicit whole-body human experience under fixed grounding?

C ACTION INTERFACES AND TRAINING DETAILS

λ_0 uses a 2B Qwen3.5 vision–language backbone, layer-wise visual–language features, a generative action expert, and explicit embodiment conditioning. The Stage-I hand interface, Stage-II whole-body human and robot-compatible interfaces, native real-robot interface, and SIMPLE interface retain separate action projections and state contracts. All interfaces predict 50-step action chunks, with masks excluding unavailable blocks and padded dimensions from the loss.

Stage-I pre-training uses a global batch size of 1024, a base learning rate of 10^{-5} , and a 0.1 learning-rate multiplier on the vision–language backbone. An exponential moving average with decay 0.999 is updated after each optimizer step and initializes the next stage. Checkpoints contain the online parameters, EMA parameters, optimizer state, global step, data-state metadata, and random-number states required for deterministic resumption.

Stage-II supervision is constructed offline. The wearable body stream is processed by the frozen SONIC controller in a G1 MuJoCo environment, while hand articulation is retargeted to the bilateral hand interface. The robot-compatible record stores 29-DoF body state, bilateral hand state, projected gravity, the body-motion token, and bilateral hand commands. The human record preserves wrist translation and rotation, wrist-local finger geometry, and the corresponding body-motion token. Both records point to the same first-person frames and episode timestamps but carry different action-space identifiers. Training reads only finalized LeRobot v2.1 records; hand estimation, simulation, and retargeting are never executed online.

Stage III jointly samples the converted 600 task-aligned human demonstrations and native robot episodes. The two streams preserve distinct state-action interfaces and validity masks while updating the shared backbone and action expert. Checkpoints record both dataset versions, the fixed human-robot sampling ratio, per-stream sample counts, and optimizer state so that resuming does not alter the training mixture.

With mixture weights λ_H and λ_R , Stage III minimizes

$$\mathcal{L}_{\text{post}} = \lambda_H \mathbb{E}_{i \sim \mathcal{D}_H^T} [\mathcal{L}_i] + \lambda_R \mathbb{E}_{i \sim \mathcal{D}_R^e} [\mathcal{L}_i], \quad (6)$$

where each $\mathcal{L}_i$ uses the valid-target masked MSE in the main text. Mixture weights are selected before the main evaluation and remain fixed for every HUMANVERSE-500 scaling condition.

C.1 TRAJECTORY DEFINITIONS AND TRANSFER PARADIGM

A training domain d provides trajectories

$$\tau_i^d = \left\{ (\mathbf{x}_{i,t}^d, \mathbf{s}_{i,t}^d, \ell_i^d, \mathbf{a}_{i,t+1:t+H}^d, \mathbf{m}_{i,t+1:t+H}^d) \right\}_{t=0}^{T_i-1}, \quad H = 50, \quad (7)$$

where $\mathbf{x}_{i,t}^d$ is the egocentric RGB observation, $\mathbf{s}_{i,t}^d$ is the current state, ℓ_i^d is the language instruction, $\mathbf{a}_{i,t+1:t+H}^d$ is a future action chunk, and $\mathbf{m}_{i,t+1:t+H}^d$ marks its valid temporal and feature dimensions. Each domain is associated with an embodiment identifier $e(d)$ and an action space $q(d)$; neither identifier implies numerical correspondence between domains.

We distinguish four data collections. $\mathcal{D}_E$ is the union of seven public egocentric hand-object datasets, $\mathcal{D}_W$ is the general whole-body corpus HUMANVERSE-500, $\mathcal{D}_T$ contains the separate 600 task-aligned human demonstrations, and $\mathcal{D}_R^e$ contains native demonstrations from target robot e . From each trajectory in $\mathcal{D}_W$, we construct a human view $\Phi_H(\tau)$ and a robot-conditioned view $\Phi_R(\tau)$. The latter is derived from human motion with frozen retargeting and simulation modules; it is not a robot demonstration. Our transfer paradigm is therefore

$$\theta^{(0)} \xrightarrow{\mathcal{D}_E} \theta^{(1)} \xrightarrow{\Phi_H(\mathcal{D}_W) \cup \Phi_R(\mathcal{D}_W)} \theta^{(2)} \xrightarrow{\mathcal{D}_T \cup \mathcal{D}_R^e} \theta_e^{(3)}. \quad (8)$$

Stage I learns interaction geometry, Stage II couples manipulation with whole-body motion, and Stage III grounds the transferred representation in a task and a physical embodiment.

C.2 ACTION DENOISING AND VALID-TARGET NORMALIZATION

A common learning objective. The checkpoints used for the main evaluation predict the clean action x_0 . For each sample, we draw $u \sim \mathcal{U}[0, 1)$ and set $\sigma = 0.001 + 0.999u^{2/3}$, corresponding to a rescaled Beta(1.5, 1) distribution. Given $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we construct

$$\tilde{\mathbf{a}}_{h,\sigma} = (1 - \sigma_h) \mathbf{a}_h + \sigma_h \epsilon_h, \quad \hat{\mathbf{a}} = f_{\theta}(\mathbf{x}, \ell, \mathbf{s}, e, q, \tilde{\mathbf{a}}_{\sigma}, \sigma). \quad (9)$$

For RTC training, half of the samples use no action prefix. For the remainder, we sample up to eight valid prefix steps, set their noise level to zero, and exclude them from supervision. These clean actions condition prediction of the remaining suffix.

Let M contain every valid, non-prefix scalar target (i, h, j) in a batch. The implemented objective is

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{\sigma, \epsilon, r} \left[\frac{1}{|M|} \sum_{(i,h,j) \in M} (\hat{a}_{i,h,j} - a_{i,h,j})^2 \right]. \quad (10)$$

The validity mask removes unavailable dimensions, missing-hand annotations, and cross-interface padding. Action-space blocks are used to report diagnostic losses, but the optimizer averages all valid scalar targets without block reweighting. The mid-training and post-training recipes draw four independent noise realizations per observation before applying the same masked objective.

C.3 STAGE-SPECIFIC TRAINING OBJECTIVES

All stages use the conditional denoising objective above, but differ in the data distribution, prediction interface, and role of supervision. Stage transitions initialize from the preceding exponential-moving-average (EMA) checkpoint, preserving learned representations while activating the interfaces needed by the next stage.

Stage I: interaction-prior pre-training. This stage learns visual and language-conditioned hand-object interaction structure before introducing whole-body targets. We initialize the shared backbone from a vision-language checkpoint and optimize it jointly with the action expert and egocentric hand interface on EgoDex, EgoVerse, HO-Cap, HOI4D, HoloAssist, HOT3D, and VITRA-1M (Hoque et al., 2025; Punamiya et al., 2026; Wang et al., 2025; Liu et al., 2022; Wang et al., 2023b; Banerjee et al., 2025; Li et al., 2026a). VITRA-1M additionally indexes activity from Ego4D, EPIC-KITCHENS, Ego-Exo4D, and Something-Something V2 (Grauman et al., 2022; Damen et al., 2022; Grauman et al., 2024; Goyal et al., 2017). If N_d is the number of usable frames in source d , square-root sampling gives

$$p_1(d) = \frac{\sqrt{N_d}}{\sum_{d' \in S_E} \sqrt{N_{d'}}}, \quad \theta^{(1)} = \arg \min_{\theta} \mathbb{E}_{d \sim p_1, \tau \sim \mathcal{D}_d, t} [\mathcal{L}_{\text{denoise}}^E(\tau, t)]. \quad (11)$$

The prediction target is future relative wrist motion and local finger geometry, not humanoid control. Square-root sampling retains coverage from small sources while preventing the largest corpus from setting the entire training distribution. Dataset identity remains available to the conditioning module, and all supervision retains its native frame index, provenance, and validity mask.

Stage II: whole-body motion-prior mid-training. Stage I does not explicitly supervise the body motion that brings the hands to an object or transports it after grasping. Starting from its EMA checkpoint, Stage II introduces synchronized body and hand targets from HUMANVERSE-500. This stage is intended to connect the interaction prior to whole-body coordination before task-specific robot adaptation. For every HUMANVERSE-500 trajectory, we preserve the same RGB, instruction, episode identity, and timestamp support in both views. The human view predicts body-motion tokens and finger geometry (Appendix C.5). To obtain the paired robot view, the frozen SONIC motion encoder converts tracked body windows into $\mathbf{z}_{1:T}$; a frozen SONIC controller then executes these tokens in a free-base G1 MuJoCo model, and deterministic hand retargeting maps reconstructed MANO-21 geometry to Revo2 commands (Luo et al., 2025). In shorthand,

$$\mathbf{z}_t = \mathcal{E}_{\text{motion}}(\mathbf{b}_{t:t+10}), \quad (\mathbf{q}_t, \mathbf{g}_t^{\text{proj}}) = \text{Rollout}_{\text{G1}}(\mathbf{z}_{1:t}), \quad \mathbf{u}_t^x = \mathcal{R}_{\text{hand}}(\mathbf{J}_t^x). \quad (12)$$

These operators are frozen: Stage II learns from their outputs but does not back-propagate through data conversion. We sample the human and robot views with equal probability; within each view, collection d is sampled in proportion to its usable frames, followed by uniform episode and timestep sampling. Hence

$$\mathcal{L}_2(\theta) = \frac{1}{2} \mathbb{E}_{\Phi_H(\mathcal{D}_W)} [\mathcal{L}_{\text{denoise}}^H] + \frac{1}{2} \mathbb{E}_{\Phi_R(\mathcal{D}_W)} [\mathcal{L}_{\text{denoise}}^R], \quad \theta^{(2)} = \arg \min_{\theta \leftarrow \text{EMA}(\theta^{(1)})} \mathcal{L}_2. \quad (13)$$

The Stage-I hand head is retained but frozen; the shared backbone, action expert, and active whole-body interfaces are trained on these two views. Both views originate from human recordings: the simulated robot-compatible view is not a real-robot demonstration and does not establish real-world task success. Neither the 600 task-aligned human demonstrations nor native robot demonstrations enter this stage. The resulting EMA checkpoint initializes Stage III.

Stage III: embodiment-grounded post-training. We initialize from the Stage-II exponential-moving-average checkpoint and adapt to target embodiment e . This stage addresses the remaining

gap between motion supervision and execution in the target robot’s physical environment. Task-aligned human demonstrations preserve interaction and object coverage through the human whole-body interface; native robot trajectories provide executable state transitions through the robot interface. With a fixed human sampling coefficient α , the objective is

$$\mathcal{L}_3^e(\theta) = \alpha \mathbb{E}_{\mathcal{D}_T}[\mathcal{L}_{\text{denoise}}^H] + (1 - \alpha) \mathbb{E}_{\mathcal{D}_R^e}[\mathcal{L}_{\text{denoise}}^R], \quad \theta_e^{(3)} = \arg \min_{\theta \leftarrow \text{EMA}(\theta^{(2)})} \mathcal{L}_3^e(\theta). \quad (14)$$

The shared backbone, action expert, embodiment conditioning, and active interfaces are optimized jointly. We hold α , the Stage-III update budget, and $\mathcal{D}_R^e$ fixed in the human-data scaling comparison; robot-data-efficiency experiments vary only $\mathcal{D}_R^e$. The output is a task- and embodiment-adapted policy, rather than direct deployment of a human-only model. In the default 2B recipe, Stages I–III use 200k, 100k, and 50k optimizer updates, respectively, with the same 50-step prediction horizon.

C.4 PHYSICAL GROUNDING AND DEPLOYMENT

Fig. 4 shows the target humanoid and teleoperation setup used to collect $\mathcal{D}_R^e$ for Stage III. Unlike the simulated view in Stage II, these demonstrations pair observations and robot states with actions recorded during physical task execution. They provide the embodiment-specific supervision that complements the task-aligned human demonstrations.

At deployment, the policy is conditioned on the target robot’s observation, instruction, state, and embodiment identifier, and uses only its robot output interface. For G1, the predicted body-motion tokens are interpreted by the SONIC control interface and paired with robot hand commands; reconstructed human joints are not sent directly to actuators. Human streams are training resources, not required deployment inputs. Other embodiments retain their native interfaces. Controller timing and task-evaluation constraints are separate from the shared action-prediction objective.

C.5 STATE–ACTION DEFINITIONS

Egocentric hand interface. For hand $\chi \in \{L, R\}$, let $\mathbf{p}_t^\chi \in \mathbb{R}^3$ and $\mathbf{R}_t^\chi \in \text{SO}(3)$ denote the wrist pose, let $\rho_6(\mathbf{R}) \in \mathbb{R}^6$ be its continuous 6D representation, and let $\mathbf{g}_t^\chi \in \mathbb{R}^{60}$ contain the 20 non-wrist joints in a wrist-local, palm-aligned, scale-normalized frame. The current hand state is

$$\mathbf{h}_t = [\mathbf{p}_t^L, \rho_6(\mathbf{R}_t^L), \mathbf{g}_t^L, \mathbf{p}_t^R, \rho_6(\mathbf{R}_t^R), \mathbf{g}_t^R] \in \mathbb{R}^{138}. \quad (15)$$

All future wrists are expressed in the camera frame at the query time t . Stage-I actions retain relative wrist motion and absolute local finger shape:

$$\mathbf{a}_{t+h}^{E,\chi} = [\mathbf{p}_{t+h}^\chi - \mathbf{p}_t^\chi, \rho_6((\mathbf{R}_t^\chi)^\top \mathbf{R}_{t+h}^\chi), \mathbf{g}_{t+h}^\chi], \quad \mathbf{a}_{t+h}^E = [\mathbf{a}_{t+h}^{E,L}, \mathbf{a}_{t+h}^{E,R}] \in \mathbb{R}^{138}. \quad (16)$$

This construction removes arbitrary camera-world offsets while retaining the geometry needed to distinguish reaching, grasping, and in-hand interaction.

Paired whole-body interfaces. Let $\mathbf{z}_t \in \mathbb{R}^{64}$ be the motion token produced by the frozen SONIC encoder. The human view uses the current bilateral hand state but predicts future body motion and finger geometry,

$$\mathbf{s}_t^H = \mathbf{h}_t \in \mathbb{R}^{138}, \quad \mathbf{a}_{t+h}^H = [\mathbf{z}_{t+h}, \mathbf{g}_{t+h}^L, \mathbf{g}_{t+h}^R] \in \mathbb{R}^{184}. \quad (17)$$

For the robot-conditioned view, a closed-loop free-base G1 rollout provides 29 body-joint positions and projected gravity $\mathbf{g}_t^{\text{proj}} \in \mathbb{R}^3$, while hand retargeting provides six normalized Revo2 commands per side. Its interface is

$$\begin{aligned} \mathbf{s}_t^R &= [\mathbf{q}_t^{1:29}, \mathbf{u}_t^L, \mathbf{u}_t^R, \mathbf{g}_t^{\text{proj}}] \in \mathbb{R}^{44}, \\ \mathbf{a}_{t+h}^R &= [\mathbf{z}_{t+h}, \mathbf{u}_{t+h}^L, \mathbf{u}_{t+h}^R] \in \mathbb{R}^{76}. \end{aligned} \quad (18)$$

The same robot interface is used for native G1 post-training data. Other embodiments, including SIMPLE, retain their native state and actuator blocks. Variable-width states and actions are padded only at collation time; masks remove padding, episode boundaries, failed conversions, and invisible hands from supervision.

C.6 LAYER-WISE ARCHITECTURE AND CONDITIONING

λ_0 combines a shared Qwen3.5 vision–language backbone with a layer-wise generative action expert. Normalized proprioception is quantized into 256 bins and appended to the instruction. The backbone then produces hidden states $\mathbf{C}^{1:L} = V_\phi(\mathbf{x}, [\ell; Q(\mathbf{s}^q)])$. For action space q , a domain-specific encoder maps the noisy action and a sinusoidal noise embedding to action tokens,

$$\mathbf{u}_h^{q,0} = E_q([\tilde{\mathbf{a}}_h^q, \gamma(\sigma_h)]). \quad (19)$$

An embodiment-specific 32-token soft prompt and a learned embodiment bias condition the shared expert. At VLM layer l , the corresponding projected context is injected into the l th expert block,

$$(\mathbf{P}_e^l, \mathbf{U}^{q,l}) = F_\psi^l(\text{Proj}_l(\mathbf{C}^l), \mathbf{P}_e^{l-1}, \mathbf{U}^{q,l-1}), \quad \hat{\mathbf{a}}^q = D_q(\mathbf{U}^{q,L}). \quad (20)$$

Thus V_ϕ and $F_\psi^{1:L}$ transfer across stages, while (E_q, D_q) preserve the semantics and dimensionality of each state–action space. A new embodiment appends conditioning rows; if it introduces a new action contract, its interface is appended without overwriting previous ones.

C.7 BASELINE IMPLEMENTATION DETAILS

TODO: Document the baseline checkpoints, training data, adaptation settings, and inference configurations used in our evaluations.

D SCALING CONTROLS

The human-data scaling axis is expressed as a fraction of the complete HUMANVERSE-500 corpus. The 0%, 5%, 10%, 25%, 50%, and 100% conditions define the evaluation grid for Fig. 6. Nonzero subsets are nested, sampled at the episode level from the frozen training partition, and stratified by activity and recording session before chunk extraction. The Stage-I checkpoint, 600 task-aligned human demonstrations, robot post-training set, Stage-III mixture, validation set, and evaluation manifest remain fixed.

The primary compute-matched comparison uses the same number of Stage-II optimizer updates for every nonzero fraction; the 0% reference omits Stage II. The complementary exposure-matched comparison limits repeated exposure to each episode and reports the resulting compute difference. Validation loss and closed-loop metrics are analyzed separately so a reduction in the Stage-II objective is not treated as equivalent to improved robot control.

Robot-data-efficiency subsets are nested at the episode level. The 5%, 10%, 25%, and 50% sets are contained in the 100% set and stratified by task and initial-condition family. Evaluation configurations are excluded before subsampling. Each fraction compares robot-only post-training, Stage-III human–robot training without Stage II, Stage-II mid-training followed by robot-only post-training, and λ_0 . Update counts and stopping rules are matched, and the 600 task-aligned demonstrations remain fixed whenever that stream is enabled.

D.1 LOGGED VALIDATION RUNS AND PLOTTING PROTOCOL

Fig. 5 reports the recorded `id2.lib` runs from the `Lambda0-Pico-Scaling` project at 5%, 10%, 25%, 50%, and 100% training-data fractions. Each run uses the 2B variant, training seed 43, and 100,000 optimization steps. We report the logged total validation loss (`loss_total`) on the clean and representative validation sets, evaluated every 1,000 steps with validation seed 20260813. Data fractions are reported directly rather than converted to hours. These are offline prediction metrics, not closed-loop robot success rates.

The two validation sets probe different data-quality regimes. *Clean* contains held-out capture sessions selected for high frame-weighted validity of both SONIC whole-body action annotations and bilateral hand annotations: both session-level validity ratios must reach the respective upper-quartile thresholds in the original split. Thus, “clean” refers to annotation validity, not visual appearance or task difficulty. *Representative* is a separate holdout stratified by data source and quartile bins of the two validity ratios, covering a broader range of annotation quality. The frozen sets contain 1,008

Table 6: Scaling HUMANVERSE-500 under fixed downstream supervision. The 600 task-aligned human demonstrations, robot demonstrations, Stage-III mixture, and model capacity are held constant across rows. Values are reported as mean $\pm$ standard deviation over three training seeds; real-robot metrics additionally include paired-rollout confidence intervals. Pending measurements are indicated by dashes.

HumanVerse-500 subset (%)	Human val. loss	SIMPLE success (%)	Real-robot success (%)	Real-robot progress (%)	Held-out progress (%)
0	—	—	—	—	—
5	—	—	—	—	—
10	—	—	—	—	—
25	—	—	—	—	—
50	—	—	—	—	—
100	—	—	—	—	—

and 513 episodes, respectively, each from 44 capture sessions. Their sessions are mutually disjoint and excluded from training; both sets remain fixed across all five training-data fractions.

Following the diagnostic decomposition used by EgoScale (Zheng et al., 2026b), we distinguish validation dynamics from the best recorded validation error at each data scale. The best validation loss on both holdouts decreases across the five data scales. We show measured minima with descriptive log-linear fits; these are not evidence of a universal scaling law. Downstream transfer is evaluated separately.

The two plots in panel (a) show the 100 unsmoothed evaluations from step 1,000 through 100,000 on a shared logarithmic loss axis spanning 0.005–0.02 and a step axis starting at zero. Step 0 is omitted from these panels for readability but retained in the source data. Panel (b) selects each holdout’s minimum independently over the full 0–100,000-step window; minima need not occur at the same step. Solid circles denote clean and pale-gray-centered circles representative validation. Each data fraction has one run, so no across-seed error bars are shown.

For each holdout, we fit $\hat{L}(D) = a + b \ln D$ by unweighted ordinary least squares to the five measured minima, where $D \in \{5, 10, 25, 50, 100\}$ is the training-data percentage. Clean gives $a = 0.0112283$, $b = -0.00126478$, and $R^2 = 0.9901$; representative gives $a = 0.0129477$, $b = -0.00129865$, and $R^2 = 0.9868$. Pale-purple dashed lines show these fits only within the observed range. Their slope magnitudes differ by approximately 2.7% relative to clean. A doubling of data predicts an absolute loss reduction of $-b \ln 2$: 0.000900 for representative and 0.000877 for clean. Thus, the two holdouts exhibit similar absolute, not necessarily relative, improvements over the observed data range. With one run per scale, this is a descriptive comparison, not a statistical test of equal slopes. The reported R^2 values measure in-sample fit, not cross-seed reproducibility or extrapolation accuracy.

Table 7: Robot-data efficiency under nested downstream demonstration subsets. Each entry reports macro-averaged success / weighted task progress (%) across the four real-world tasks. Pending evaluations are indicated by dashes.

Robot data fraction	Robot-only post-training	Stage III only human + robot	Stage II + robot post-train	λ_0
5%	– / –	– / –	– / –	– / –
10%	– / –	– / –	– / –	– / –
25%	– / –	– / –	– / –	– / –
50%	– / –	– / –	– / –	– / –
100%	– / –	– / –	– / –	– / –

E SIMULATION EVALUATION PROTOCOL

We separate the six core leaderboard tasks from six additional SIMPLE tasks. Each task has ten rollouts at each of Levels 0, 1, and 2. An entry $a/b/c$ below denotes success rates in percent at these three levels; with ten trials, each success contributes ten percentage points. For successful-trial count $n_{k,l}$ on task k at level l , the main-table metrics are $S_l = 100 \sum_{k=1}^6 n_{k,l}/60$ and $S_{\text{overall}} = (S_0 + S_1 + S_2)/3$. λ_0 uses single-task training. StarVLA and λ_0 entries are from our evaluations; public baseline entries retain their source protocols.

The current λ_0 summary selects the best recorded configuration separately for each task, rather than a shared checkpoint and inference setting. Handover, BendPick, and XMovePick use the velocity-field variant at 50k steps. Grasp uses the legacy variant at 100k steps. Mobile P&P uses the legacy variant at 100k steps with RTC enabled. XMoveBendPick uses the legacy variant, with tied results at 70k and 90k steps. RTC settings for the other tasks were not specified in the latest result record. The six tasks sum to 162 successes in 180 trials. Configuration selection was not established here as validation-only; these numbers should not be interpreted as a fixed-configuration comparison. The Full Recipe row in Table 3 uses this updated task-wise summary (90.0%, 93.3%, and 86.7% at Levels 0–2; 90.0% overall); matched checkpoint and inference selection across ablations remains to be verified.

E.1 DETAILED RESULTS ON THE SIX LEADERBOARD TASKS

Table 8 reports the task-level rates underlying Table 2. Public baseline counts follow the official leaderboard accessed September 13, 2026, including its updated ACT results. Mobile P&P corresponds to locomotion pick between tables.

Table 8: Detailed success rates (%) on the six SIMPLE leaderboard tasks. Each entry gives Level 0 / Level 1 / Level 2, with ten trials per level. Public baselines follow the official leaderboard; StarVLA and λ_0 are evaluated by us.

Model	Task					
	XMovePick	BendPick	Handover	Mobile P&P	Grasp	XMoveBendPick
ACT (Zhao et al., 2023)	100/100/50	100/90/90	70/70/100	50/50/50	100/100/80	90/100/100
Diffusion Policy (Chi et al., 2025)	30/30/20	100/80/60	30/20/40	40/0/0	80/90/80	0/0/0
EgoVLA (Yang et al., 2025)	0/10/20	70/50/80	0/40/30	0/0/0	100/100/70	30/50/40
DreamZero (Ye et al., 2026)	100/100/100	90/90/80	70/80/90	50/30/30	90/100/70	0/0/10
H-RDT (Bi et al., 2025)	0/0/20	0/0/10	0/10/0	0/0/0	0/0/0	0/0/0
InternVLA-M1 (Chen et al., 2025)	0/0/0	50/50/0	0/0/0	0/0/0	0/0/0	30/50/70
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	70/50/10	100/100/80	50/40/50	30/30/30	100/100/80	0/0/0
GR00T N1.6 (GEAR Team et al., 2025)	100/100/70	70/70/60	10/30/30	0/0/0	90/90/70	40/40/10
Ψ_0 (Wei et al., 2026b)	100/100/60	100/100/100	70/70/100	70/50/60	100/100/80	100/90/90
StarVLA (StarVLA Community, 2026)	100/90/90	70/90/70	80/80/100	90/80/80	90/100/70	70/100/100
λ_0 (Ours)	100/100/80	100/100/100	90/100/100	90/70/70	100/90/80	60/100/90

E.2 SUCCESS RATES ON SIX ADDITIONAL TASKS

Table 9 reports tasks outside the leaderboard aggregate. Over these six tasks and three levels, overall success is 70.6% for Ψ_0 , 78.9% for StarVLA, and 73.3% for λ_0 .

Table 9: Success rates (%) on six additional SIMPLE tasks, excluded from the main leaderboard aggregate. Each entry gives Level 0 / Level 1 / Level 2, with ten trials per level. Published Ψ_0 results follow Wei et al. (2026a); StarVLA and λ_0 are evaluated by us.

Model	Task					
	Close Door	Open Oven	Open Faucet	Pick Place Hug Container	Push Office Chair	Open Trash Can
Ψ_0 (Wei et al., 2026b)	100/100/100	70/50/40	30/30/40	70/60/30	100/100/100	70/90/90
StarVLA (StarVLA Community, 2026)	100/100/100	90/80/100	50/70/60	70/60/80	100/80/80	50/70/80
λ_0 (Ours)	100/100/100	80/100/80	70/40/30	40/40/30	100/100/100	80/40/90

F REAL-WORLD EVALUATION PROTOCOL

For each reported policy, each of the four tasks is evaluated with ten frozen initial configurations. Training uses three seeds. Configuration identities, object instances, target regions, task instructions, timeouts, and terminal predicates are fixed before evaluating any policy. The following stage-completion criteria and terminal predicates are shared by all methods.

F.1 WEIGHTED TASK PROGRESS

For task t with K_t stages, the evaluator assigns $s_{t,k} \in \{0, 1\}$ to stage k : a completed stage receives its full preassigned point value $w_{t,k}$, and an uncompleted stage receives zero. The point value is the weight, not a separate subjective score. The weights satisfy $w_{t,k} > 0$ and $\sum_{k=1}^{K_t} w_{t,k} = 100$. For rollout r , we report

$$P_t^{(r)} = \sum_{k=1}^{K_t} w_{t,k} s_{t,k}^{(r)} \in [0, 100]. \quad (21)$$

Thus, 75 points is reported as 75% progress. This is a weighted sum of binary stage outcomes, not an equal-weight stage count or a subjective continuous quality rating. A stage earns credit only when its own observable criterion is met; the sum does not additionally apply a completed-prefix gate. Repeated completion within one rollout does not earn additional points. All stages must be adjudicated before a record is considered complete; a fully adjudicated record is not necessarily a successful rollout.

For R_t rollouts per task and T evaluated tasks, task means and the macro-average are

$$\bar{P}_t = \frac{1}{R_t} \sum_{r=1}^{R_t} P_t^{(r)}, \quad \bar{P} = \frac{1}{T} \sum_{t=1}^T \bar{P}_t. \quad (22)$$

Each task therefore has equal weight regardless of its number of stages. The same stage definitions and point allocations must be fixed before evaluation and used for every method and data-scale condition. Table 10 lists every stage and its point value; the point values sum to 100 for each task. Bottle disposal has six task-execution stages and a separate five-point post-disposal stability stage.

F.2 STAGE CRITERIA AND AUTONOMOUS SUCCESS

Table 10 summarizes the task stages. For bottle disposal, approaching the bottle means bringing the hand into its effective grasp region; a stable grasp requires reliable bottle retention, and lifting requires the bottle to clear the table. Arrival at the bin requires a pose from which the robot can both operate the pedal and deliver the bottle. Pedal opening is a separate stage: touching the pedal is insufficient; the lid must open far enough to admit the bottle. The disposal stage is worth 20 points and requires release followed by the bottle entering the bin; release alone is insufficient. The final five-point stage requires the robot to remain stable for three seconds after disposal, without human

support or unintended collision. Standing still without disposing of the bottle does not satisfy this post-disposal criterion.

For toy storage, contact is distinct from a stable grasp, lifting requires table clearance, and transport requires the toy to reach the box opening. Placement and release are distinct: the toy must enter the box and remain there after the hand lets go. For bimanual box transport, both hands must establish contact, the box must remain stable relative to the grasp, and its base must clear the table before transport. Placement requires support by the target surface, followed by stable release. For chair placement, turning and alignment, bilateral contact, stable grasp and push initiation, sustained pushing, final insertion, and safe release are assessed separately.

Full-task success is a separate binary outcome, requiring the task-specific terminal predicate to hold for three seconds without human intervention. Safety and intervention are recorded separately for all tasks, even when a task includes an explicitly scored stability stage. The bottle task awards five points for post-disposal stability; the chair task assigns 20 points to final insertion and five points to release and safe termination. A progress score alone must not be used to infer autonomous success.

Table 10: Operational definitions for the real-world evaluation. Progress is the sum of point values for completed stages (shown in parentheses, totaling 100 per task); terminal success additionally requires the task-specific predicate to remain satisfied for three seconds.

Task	Scored task stages (points)	Terminal success predicate
Bottle disposal	1. Approach the bottle (10) 2. Grasp stably (15) 3. Lift clear of the table (15) 4. Arrive at the bin (15) 5. Press the pedal to open the lid (20) 6. Release the bottle into the bin (20) 7. Remain stable after disposal (5)	Bottle fully inside the trash can and robot standing without contact assistance
Single-hand toy storage	1. Approach the toy (10) 2. Establish contact (15) 3. Grasp stably (20) 4. Lift clear of the table (15) 5. Transport above the box (15) 6. Place inside the box (20) 7. Release with the toy remaining inside (5)	Toy fully inside the box, released, and not displaced during hand withdrawal
Bimanual parts-box transport	1. Approach the box (10) 2. Establish bilateral contact (15) 3. Establish a stable grasp (20) 4. Lift clear of the table (15) 5. Transport above the target surface (20) 6. Place on the target surface (15) 7. Release with the box remaining stable (5)	Box inside the target region, both hands released, and no contained parts spilled
Chair placement	1. Approach the chair (10) 2. Turn and align with the chair (15) 3. Establish bilateral contact (10) 4. Grasp stably and initiate pushing (20) 5. Push continuously toward the table (20) 6. Complete insertion under the table (20) 7. Release and stop safely (5)	Chair footprint, position, and yaw satisfy pre-registered under-table tolerances, with the robot released and stable

Main-paper Table 1 reports the paired evaluation against external baselines, and Table 3 reports the primary matched ablations under the same protocol.

G TRIAL ADJUDICATION AND STATISTICAL ANALYSIS

Each evaluation configuration receives a persistent identifier shared by all methods. Configuration order is randomized independently for each policy. Trial videos and state logs retain configuration, model, and seed identifiers, while model identity is hidden from manual adjudicators. Human contact with the robot, object, or environment after policy execution begins is an intervention and makes the rollout ineligible for autonomous success. Assisted continuations may be retained for diagnosis but are reported separately.

We use Wilson intervals for binomial full-task and milestone success. Weighted task progress and paired model differences use a stratified bootstrap that first resamples training seeds and then paired configurations within each seed. Macro-averages weight tasks equally, independent of task dura-

tion. Scaling plots show all three seed values in addition to means and intervals; regression fits are descriptive only over the measured range.

H DATA ABLATION PROTOCOL

The ablation study varies only the four principal data sources while holding the architecture, action interfaces, robot demonstrations, evaluation manifest, and per-stage optimization settings fixed. The robot-only condition removes all human streams. The remaining matched variants remove Stage-I heterogeneous ego data, Stage-II HUMANVERSE-500, or the 600 task-aligned Stage-III human demonstrations one at a time. No action-representation or architecture ablation is included. Main-paper Table 3 reports the robot-only, w/o Stage I, w/o Stage II, and full-recipe conditions on the four real-world tasks, with ten trials per task. The corresponding experiment records are `Lambda0_base`, `Lambda0_wo1`, `Lambda0_wo2`, and `Lambda0`. Their total success counts are 6/40, 20/40, 9/40, and 25/40, respectively; progress is averaged equally across the four tasks.

I OBJECT-EXPOSURE GENERALIZATION PROTOCOL

The protocol covers all four real-world tasks. Let $\mathcal{D}_A^R$ denote robot demonstrations of physical object instance A, and let $\mathcal{D}_A^H$ and $\mathcal{D}_B^H$ denote human demonstrations of A and B. No robot demonstrations of B are used. A joint-data policy is trained with $\mathcal{D}_A^R \cup \mathcal{D}_A^H \cup \mathcal{D}_B^H$; a robot-only comparison uses $\mathcal{D}_A^R$ without human-data training. *Seen* evaluates the joint-data policy on A, *Human-guided* evaluates that same policy on B, and *Zero-shot* evaluates the robot-only policy on B. Zero-shot therefore refers to object-instance transfer within a demonstrated task, not to task-level zero-shot learning or the absence of generic visual-language pretraining.

The Human-guided and Zero-shot conditions use the same B instances and evaluation configurations, holding the instruction, scene, target region, initial robot configuration, and success predicate fixed. Robot A data are shared across the two training recipes. Success and weighted progress follow the paired real-world protocol. This comparison tests the combined benefit of human-data training; it does not isolate the contribution of human B demonstrations from human A demonstrations or earlier human pretraining. Fig. 7 currently contains synthetic layout values rather than measured transfer results. Its solid checked and dashed object icons indicate presence and absence in robot demonstrations only; B is human-observed for Human-guided but unobserved in the robot-only recipe.

J FAILURE TAXONOMY

Table 11: Operational failure taxonomy for real-world evaluation. Each failed rollout is assigned its earliest unrecovered primary failure, while recovery attempts are recorded separately.

Category	Operational rule
Perception / ground- ing	Wrong target, receptacle, or task interpretation before purposeful contact
Locomotion	Fails to enter the manipulation workspace, loses balance, or cannot maintain a usable base pose
Reach / contact	Selects the correct target but fails to establish the intended hand or body contact
Grasp	Contact is established but the object is not secured or immediately slips before transport
Transport	A secured object is dropped, spilled, or carried along an invalid path
Placement	The object reaches the target workspace but violates the terminal position, orientation, containment, release, or stability predicate
Recovery	A recoverable error is detected but the policy fails to regrasp, reposition, or resume within the timeout
System / safety	Emergency stop, unintended collision, controller fault, or other execution failure after the policy begins acting

K QUALITATIVE ROLLOUTS AND FAILURE CASES

Qualitative comparisons use complete, continuous, time-aligned rollouts from matched configurations. Videos show the active milestone, first failure time, recovery attempt, and terminal predicate without skipping failed segments. We include representative successes and the most frequent failure category for each method rather than selecting only visually favorable examples.